

## 4<sup>th</sup> Set of Problems

1. Consider the robotic manipulators illustrated in Figures 1 through 2. Place the reference frames over the manipulators, according to the Denavit-Hartenberg methodology, and build their appropriate table of parameters.

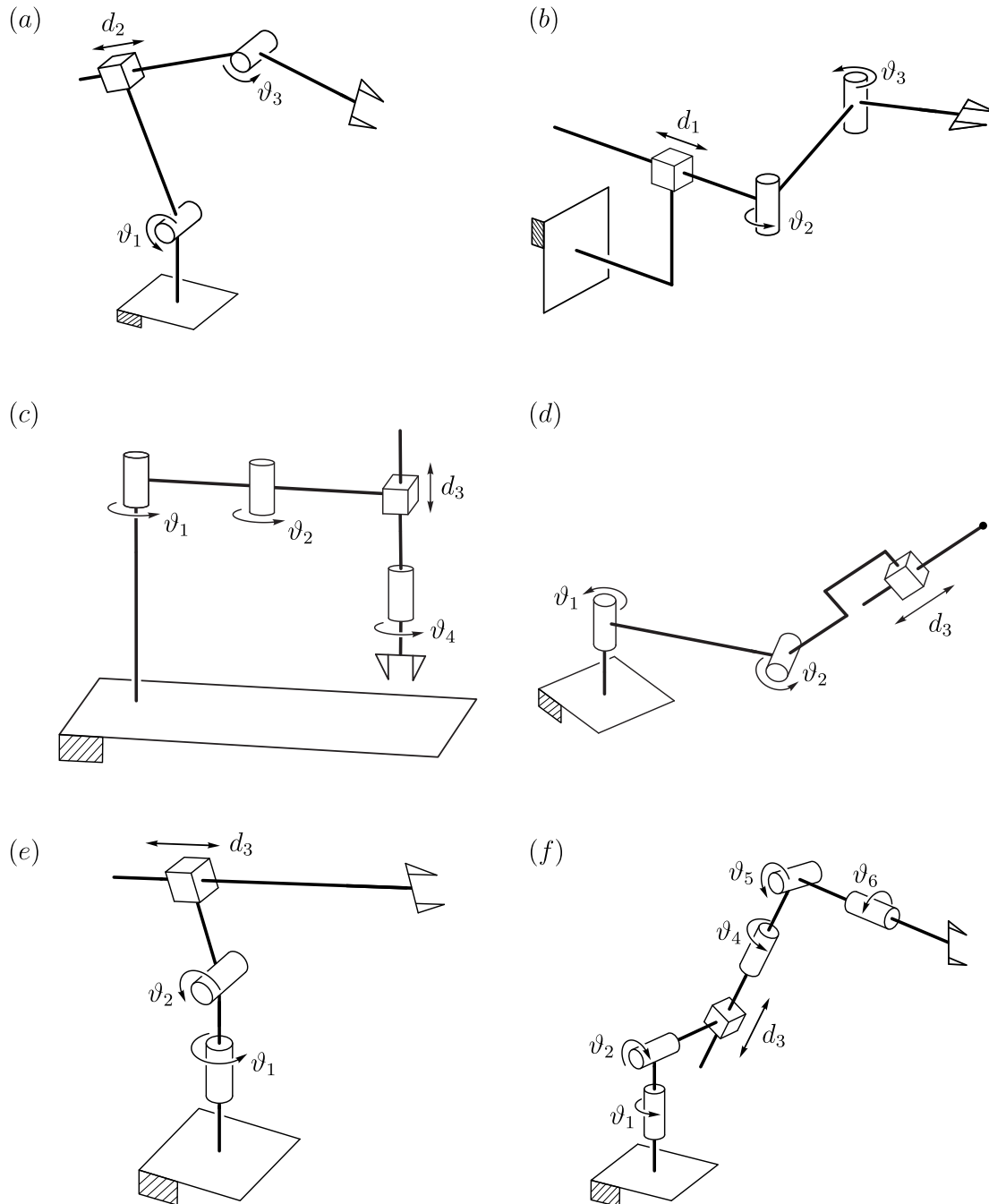


Figure 1: Manipulators. (a) Circular, (b) Stadium, (c) SCARA, (d) Toroidal, (e) Spherical and (f) Stanford.

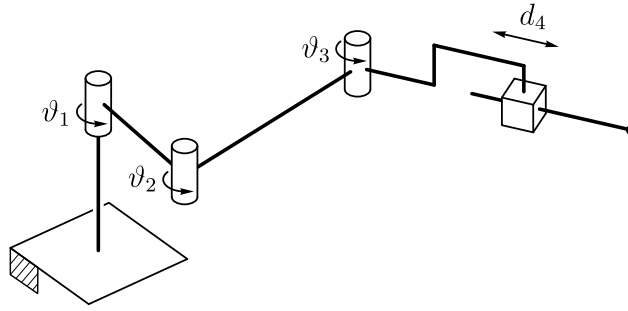


Figure 2: Piston-rod-crank system.

2. Solve the inverse kinematics for the manipulators in Figure 1.
3. Consider the kinematics of the piston-rod-crank system illustrated in Figure 2, where the fourth link is representative of the piston's translation constraint (the cylinder). Given a position and orientation for said constraint, solve the inverse kinematics of the system. As a suggestion, and since the system is redundant, you may fix either the crankshaft's rotation angle or the piston's displacement to determine the configuration of the remaining degrees of freedom.